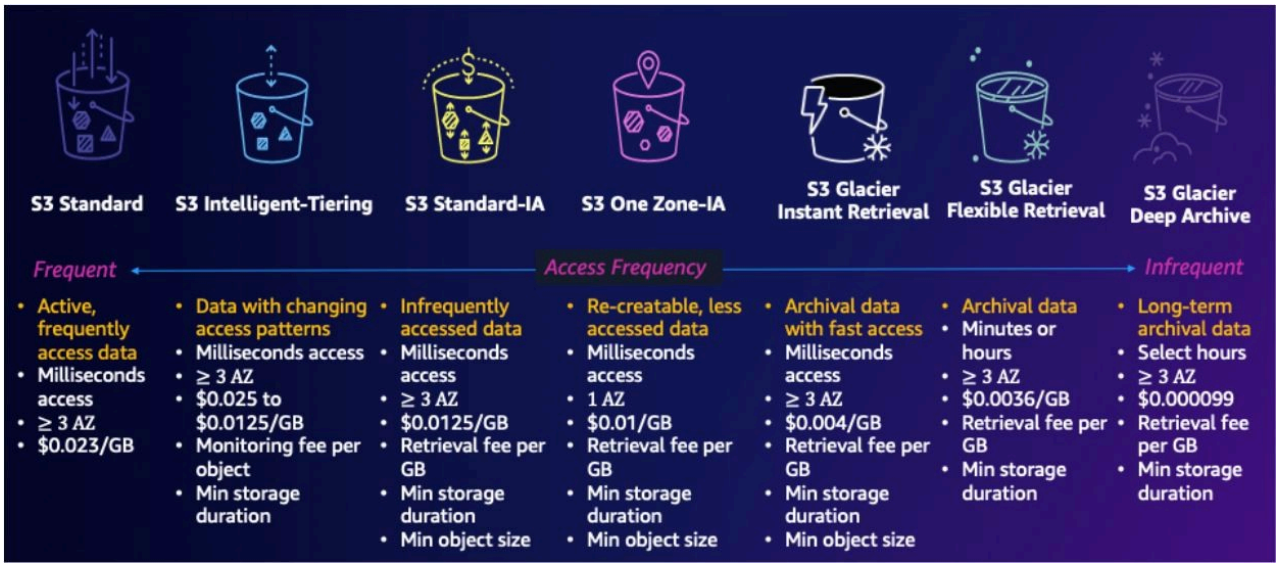
1. Tipos de almacenamiento en S3 y sus costos



(Precios aproximados en regiones estándar como us-east-1, similares en muchas regiones).

Storage Class	Precio aprox / GB mes	Acceso	Uso típico
S3 Standard	$0.023	frecuente	aplicaciones activas
S3 Intelligent-Tiering	$0.025 → $0.00099	variable	datos con acceso impredecible
S3 Standard-IA	$0.0125	ocasional	backups activos
S3 One Zone-IA	$0.01	ocasional	datos recreables
Glacier Instant Retrieval	$0.004	ocasional	archivos accesibles
Glacier Flexible Retrieval	$0.0036	raro	backups
Glacier Deep Archive	$0.00099	casi nunca	archivado a largo plazo

S3 Express One Zone	$0.11	ultra frecuente	ML / analytics alta velocidad

2. Detalle de cada tipo con ejemplos

2.1 S3 Standard

Precio:

- **$0.023 / GB / mes**

Características:

- replicación en **3 AZ**
- latencia de milisegundos
- **sin costo de retrieval**

Ejemplo:

Si almacenas:

10 TB = 10,240 GB

Costo mensual:

10,240 × 0.023 = $235.52 / mes

Uso típico:

- data lake activo
- aplicaciones web
- datasets en entrenamiento ML

2.2 S3 Intelligent-Tiering

Costo base similar a Standard pero **S3 mueve los datos automáticamente**.

Tiers posibles:

Tier	Precio

Frequent	$0.023
Infrequent	$0.0125
Archive Instant	$0.004
Archive	$0.0036
Deep Archive	$0.00099

También hay:

$0.0025 por cada 1000 objetos

para monitoreo.

Ejemplo:

Si tienes:

100 TB

y el acceso baja con el tiempo:

20 TB frecuente

30 TB infrecuente

50 TB archive

Costo:

20TB × 0.023 = $471

30TB × 0.0125 = $384

50TB × 0.004 = $204

Total ≈ $1059

Si todo estuviera en Standard:

100TB × 0.023 = $2355

Ahorro ≈ **55%**

2.3 S3 Standard-IA

Precio:

$0.0125 / GB

Pero tiene:

- mínimo **30 días**
- **costo por retrieval**

Ejemplo:

10 TB:

10,240 × 0.0125 = $128 / mes

Pero si lees mucho:

retrieval ≈ $0.01 / GB

Si lees 1 TB:

1,024 × 0.01 = $10

2.4 S3 One Zone-IA

Precio:

$0.01 / GB

Pero solo **1 AZ**.

Ejemplo:

10 TB:

10,240 × 0.01 = $102 / mes

Uso típico:

- datos recreables
- logs

2.5 Glacier Instant Retrieval

Precio:

$0.004 / GB

Acceso inmediato.

Ejemplo:

10 TB:

10,240 × 0.004 = $40 / mes

Ideal:

- data archivada pero accesible.

2.6 Glacier Flexible Retrieval

Precio:

$0.0036 / GB

Recuperación:

- minutos
- horas

Ejemplo:

10 TB

10,240 × 0.0036 = $36 / mes

2.7 Glacier Deep Archive

Precio:

$0.00099 / GB

Recuperación:

12 – 48 horas

Ejemplo:

10 TB

10,240 × 0.00099 = $10 / mes

Extremadamente barato.

3. Costos adicionales en S3 (muy importantes para la certificación)

Requests

Operación	Precio
PUT	$0.005 / 1000
GET	$0.0004 / 1000

Ejemplo:

1 millón de GET

1,000,000 / 1000 × 0.0004= $0.40

Data transfer

Entrada a S3:

GRATIS

Salida:

~ $0.09 / GB

Ejemplo:

Descargar 1 TB:

1024 × 0.09 ≈ $92

4. Costos si migras datos desde HDFS

Supongamos un **cluster Hadoop on-premise**.

Caso 1 — migración directa con DistCp

Arquitectura típica:

HDFS → DistCp → S3

Costos:

Concepto	Costo
Transferencia hacia S3	GRATIS
PUT requests	Sí
Storage	Sí

Ejemplo real: migrar 100 TB

Requests

Si tienes:

100 TB

objetos de 100 MB

Número de objetos:

100TB / 100MB

≈ 1,000,000 objetos

Costo PUT:

1,000,000 / 1000 × 0.005

= $5

Muy barato.

Storage

Si guardas en Standard:

100 TB × 0.023

≈ $2355 / mes

Si usas Glacier Instant:

100 TB × 0.004

≈ $409 / mes

Data transfer

Si migras **desde internet hacia AWS**:

$0

Pero si usas:

- **Direct Connect**
- **VPN**

pueden existir costos de red.

---

5. Ejemplo realista: Data Lake migrado de HDFS

Supongamos:

Cluster Hadoop:

1 PB

Opciones:

S3 Standard

1,000 TB × 0.023

≈ $23,000 / mes

S3 Standard-IA

1,000 TB × 0.0125

≈ $12,500 / mes

Glacier Instant

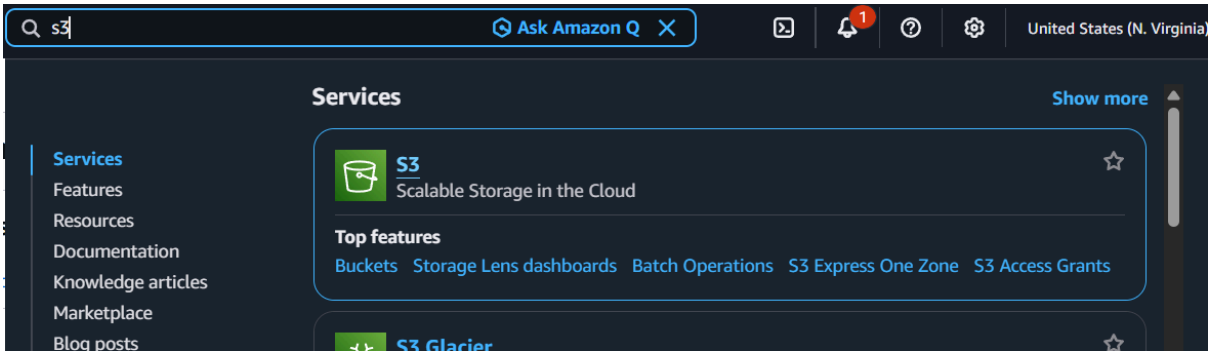
1,000 TB × 0.004

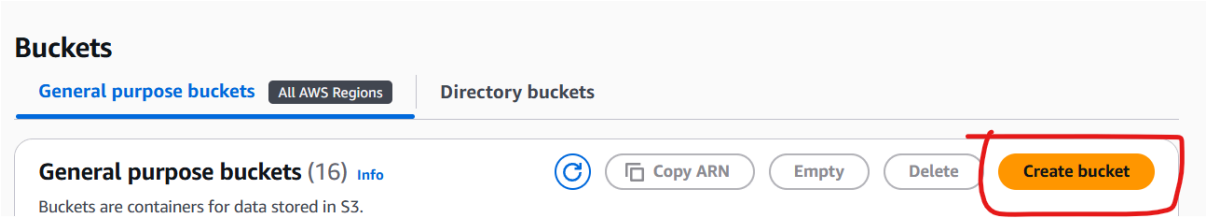
≈ $4,000 / mes

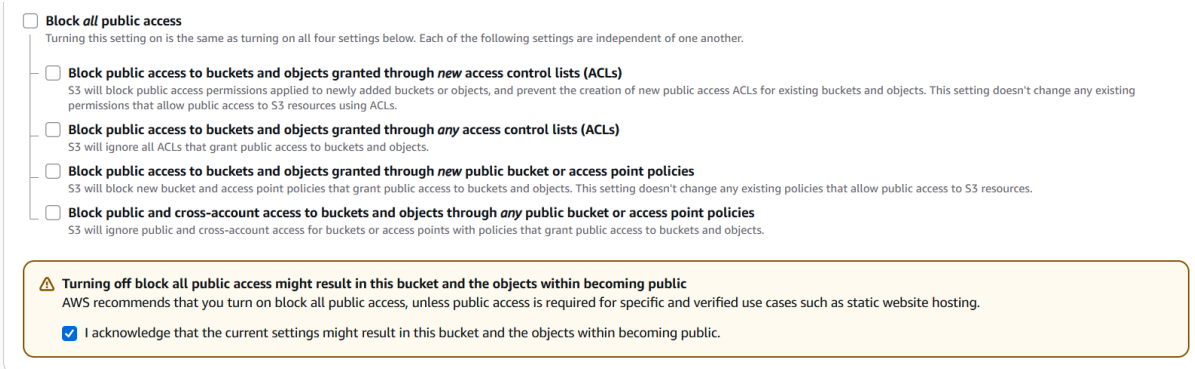
Deep Archive

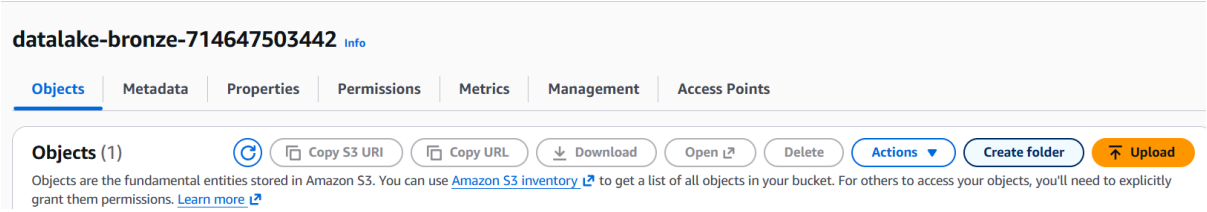
1,000 TB × 0.00099

≈ $990 / mes









Arquitectura Data Lake S3-GLUE:

```
+----------------+
|  CSV Source    |
```

```
                +--------+-------+
                         |
                         v
            +-------------------+
            |   S3 Bronze       |
            | raw CSV           |
            +---------+---------+
                      |
                      v
              Glue Job 1 (Clean)
                      |
                      v
            +-------------------+
            |   S3 Silver       |
            | parquet curated   |
            +---------+---------+
                      |
                      v
              Glue Job 2 (Transform)
                      |
                      v
            +-------------------+
            |   S3 Gold         |
            | aggregated data   |
            +-------------------+
```

---

1. Crear los Buckets

Supongamos tu cuenta `123456789`.

```
datalake-bronze-123456789
datalake-silver-123456789
datalake-gold-123456789
```

Estructura recomendada:

```
bronze
  sales/
      sales_small.csv

silver
  sales/
```

```
gold
  sales_metrics/
```

---

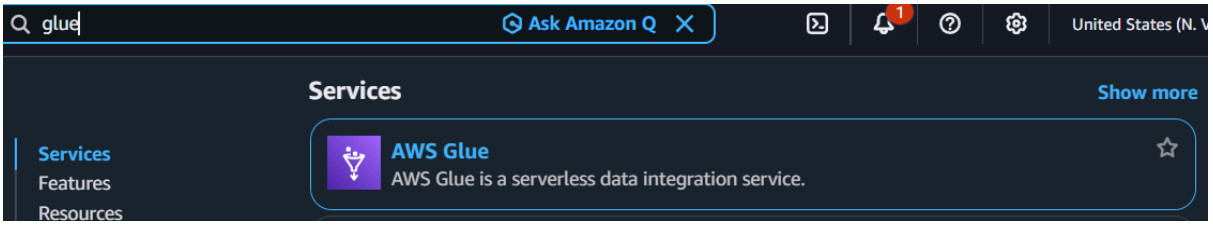
2. Dataset CSV pequeño

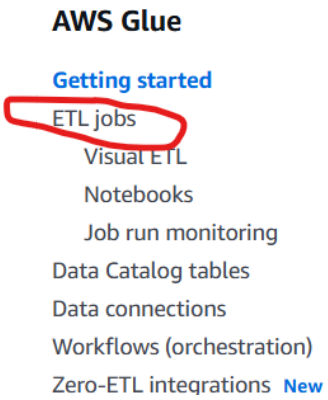
Archivo `sales_small.csv`

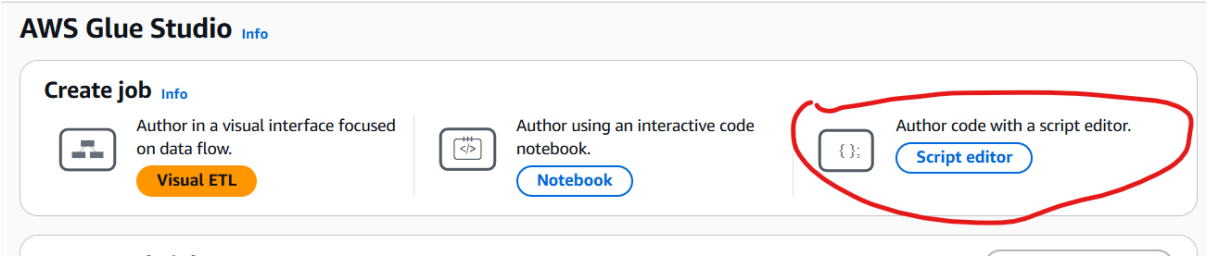
```
order_id,customer_id,product,amount,city,date
1,101,laptop,1200,Bogota,2024-01-01
2,102,mouse,25,Medellin,2024-01-01
3,101,keyboard,75,Bogota,2024-01-02
4,103,monitor,300,Cali,2024-01-03
5,104,laptop,1150,Medellin,2024-01-04
6,105,mouse,20,Bogota,2024-01-05
```

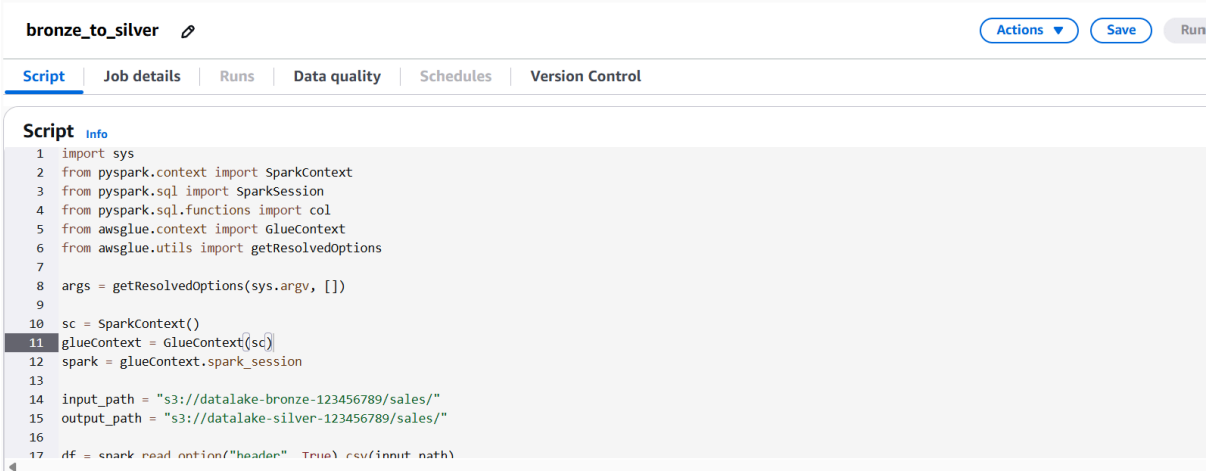
Subir a:

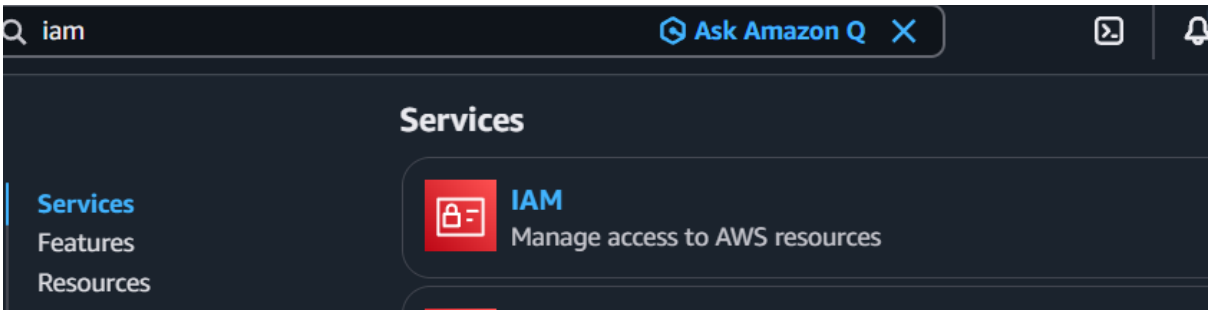
```
s3://datalake-bronze-123456789/sales/sales_small.csv
```

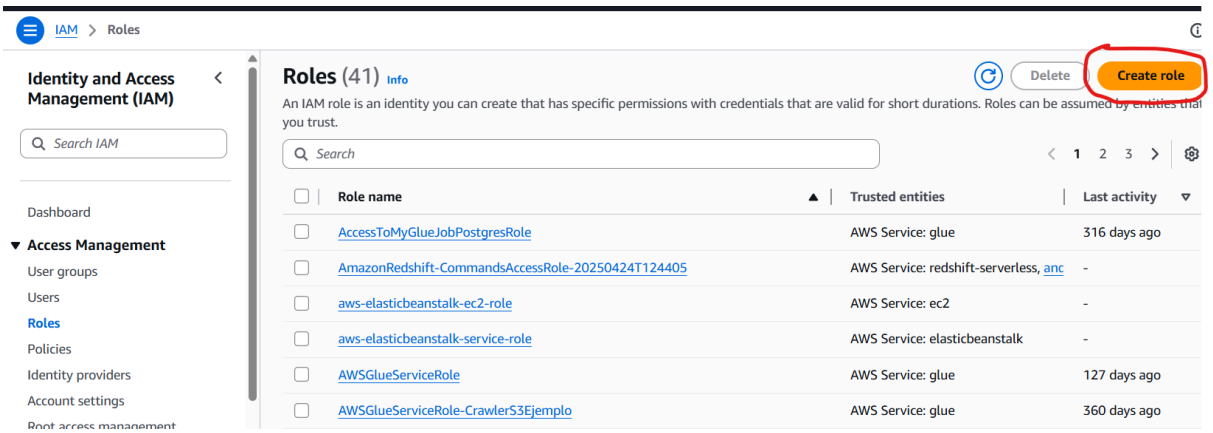





AWS Glue Studio Info
Create job Info
Author in a visual interface focused on data flow.
Visual ETL
Author using an interactive code notebook.
Notebook
Author code with a script editor.
Script editor


bronze_to_silver
Actions
Save
Run
Script
Job details
Runs
Data quality
Schedules
Version Control
Script Info
import sys
from pyspark.context import SparkContext
from pyspark.sql import SparkSession
from pyspark.sql.functions import col
from awsglue.context import GlueContext
from awsglue.utils import getResolvedOptions
args = getResolvedOptions(sys.argv, [])
sc = SparkContext()
glueContext = GlueContext(sc)
spark = glueContext.spark_session
input_path = "s3://datalake-bronze-123456789/sales/"
output_path = "s3://datalake-silver-123456789/sales/"


iam
Ask Amazon Q
Services
Services
Features
Resources
IAM
Manage access to AWS resources



3. IAM Role para Glue

Crear rol:

GlueDataLakeRole

Trust relationship:

```
{
 "Version": "2012-10-17",
 "Statement": [
   {
     "Effect": "Allow",
     "Principal": { "Service": "glue.amazonaws.com" },
     "Action": "sts:AssumeRole"
   }
 ]
}
```

Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.

Custom trust policy

Create a custom trust policy to enable others to perform actions in this account.

```
1  {
2    "Version": "2012-10-17",
3    "Statement": [
4      {
5        "Effect": "Allow",
6        "Principal": { "Service": "glue.amazonaws.com" },
7        "Action": "sts:AssumeRole"
8      }
9    ]
10 }
```

Edit statement Remove

Add actions for STS

Search actions

- All actions (sts:*)

Read

- GetAccessKeyInfo Info
- GetCallerIdentity Info
- GetFederationToken Info

IAM > Roles > Create role

+ Add new statement

10:2 JSON

+ Add new statement

Security: 0 Errors: 0 Warnings: 0 Suggestions: 0 Preview external access

Cancel Next

Step 1
Select trusted entity

Step 2
Add permissions

Step 3
Name, review, and create

Add permissions Info

Permissions policies (1/1141) Info

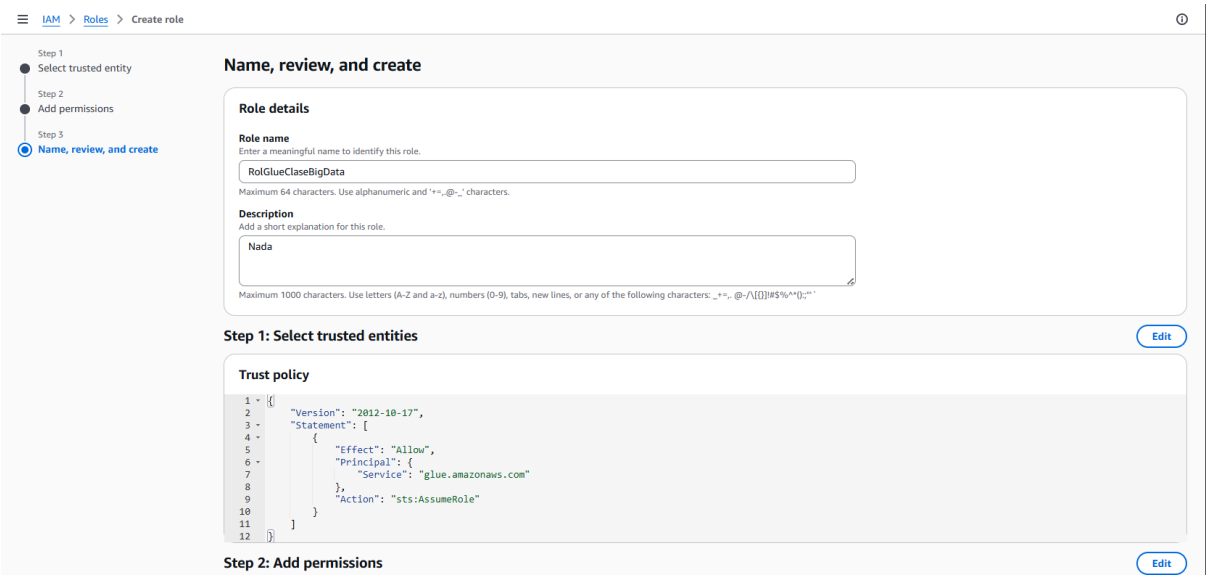
Choose one or more policies to attach to your new role.

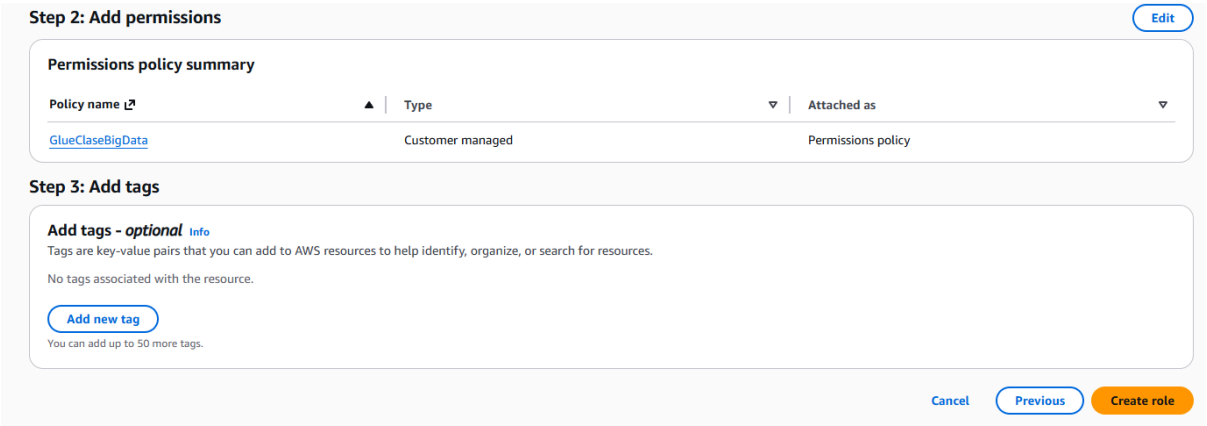
Filter by Type

clase | All types | 3 matches | 1

Policy name	Type	Description
claseIUSHPermisosEstudiantes	Customer managed	-
GlueClaseBigData	Customer managed	-
policita_roldatalakeclase	Customer managed	-

▼ Set permissions boundary - *optional*





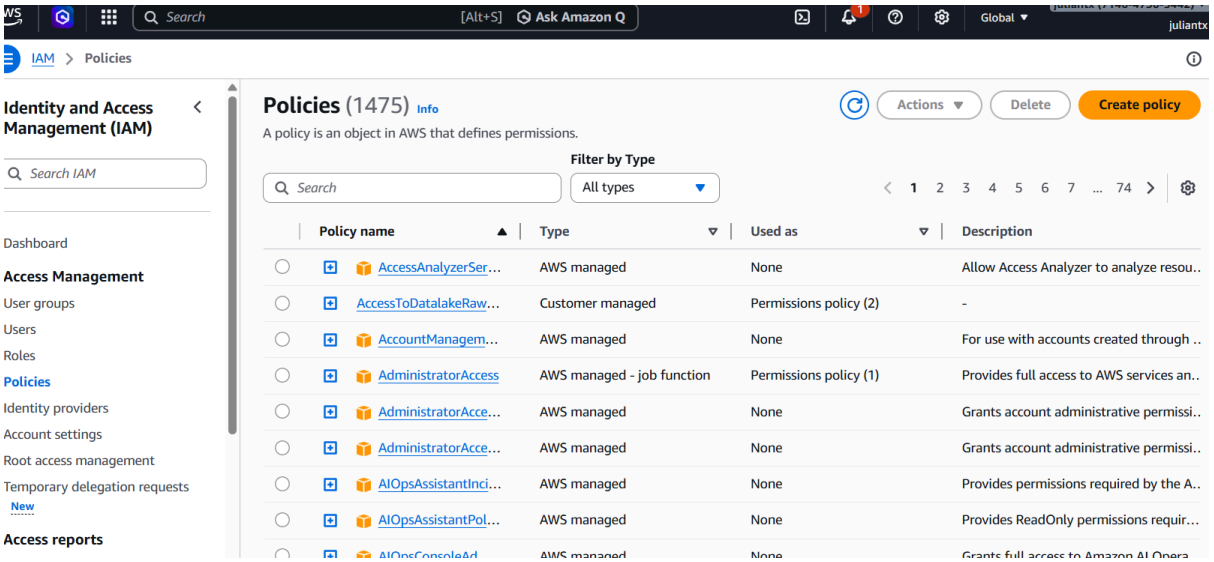
4. Policy IAM para acceso al Data Lake

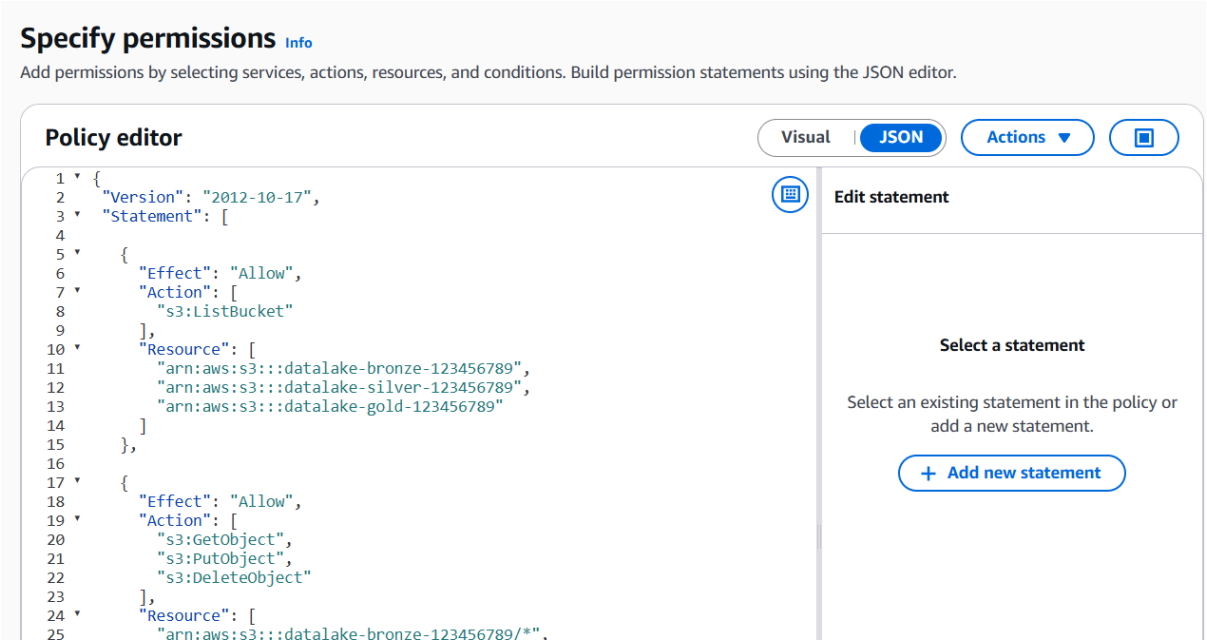
Policy:

GlueDataLakeS3Policy

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "s3:ListBucket"
            ],
            "Resource": [
```

```
                "arn:aws:s3:::datalake-bronze-714647503442",
                "arn:aws:s3:::datalake-silver-714647503442",
                "arn:aws:s3:::datalake-gold-714647503442",
                "arn:aws:s3:::aws-glue-assets-714647503442-us-east-1"
            ]
        },
        {
            "Effect": "Allow",
            "Action": [
                "s3:GetObject",
                "s3:PutObject",
                "s3:DeleteObject"
            ],
            "Resource": [
                "arn:aws:s3:::datalake-bronze-714647503442/*",
                "arn:aws:s3:::datalake-silver-714647503442/*",
                "arn:aws:s3:::datalake-gold-714647503442/*",
                "arn:aws:s3:::aws-glue-assets-714647503442-us-east-1/*"
            ]
        }
    ]
}
```





Agregar también la policy administrada:

AWSGlueServiceRole

El rol final debe tener:

```
AWSGlueServiceRole
GlueDataLakeS3Policy
```

5. Glue Job 1

```
Bronze → Silver
```

Lee CSV y guarda parquet limpio.

Archivo:

bronze_to_silver.py

```
import sys
from pyspark.context import SparkContext
from pyspark.sql import SparkSession
from pyspark.sql.functions import col
from awsglue.context import GlueContext
from awsglue.utils import getResolvedOptions

args = getResolvedOptions(sys.argv, [])
```

```python
sc = SparkContext()
glueContext = GlueContext(sc)
spark = glueContext.spark_session

input_path = "s3://datalake-bronze-123456789/sales/"
output_path = "s3://datalake-silver-123456789/sales/"

df = spark.read.option("header", True).csv(input_path)

df_clean = (
   df.withColumn("amount", col("amount").cast("double"))
)

df_clean.write.mode("overwrite").parquet(output_path)
```

Descriptions can be up to 2048 characters long.

IAM Role

Role assumed by the job with permission to access your data stores. Ensure that this role has permission to your Amazon S3 sources, targets, temporary directory, scripts, and any libraries used by the job.

RolGlueClaseBigData | Create new role

G 1X
(4vCPU and 16GB RAM)

Automatically scale the number of workers

☐ AWS Glue will optimize costs and resource usage by dynamically scaling the number of workers up and down throughout the job run. Requires Glue 3.0 or later.

Requested number of workers

The number of workers you want AWS Glue to allocate to this job.

2

OJO

Plata

Generate job insights

Script path

S3 location of the script. Path must be in the form s3://bucket/prefix/path/. It must end with a slash (/) and not include any files.

s3://aws-glue-assets-714647503442-us-east-1/scripts/ | View | Browse S3

Job metrics Info

☐ Enable the creation of CloudWatch metrics when this job runs.

Job observability metrics Info

☐ Enable the creation of additional observability CloudWatch metrics when this job runs.

Continuous logging Info

☐ Enable logs in CloudWatch.

Spark UI Info

☐ Write Spark UI logs to Amazon S3.

6. Glue Job 2

Silver → Gold

Transformación analítica.

Archivo:

silver_to_gold.py

```
import sys
from pyspark.context import SparkContext
from pyspark.sql import SparkSession
from pyspark.sql.functions import sum, count
from awsglue.context import GlueContext
from awsglue.utils import getResolvedOptions

args = getResolvedOptions(sys.argv, [])

sc = SparkContext()
glueContext = GlueContext(sc)
spark = glueContext.spark_session

input_path = "s3://datalake-silver-123456789/sales/"
output_path = "s3://datalake-gold-123456789/sales_metrics/"

df = spark.read.parquet(input_path)

df_metrics = (
   df.groupBy("city")
   .agg(
       sum("amount").alias("total_sales"),
       count("*").alias("num_orders")
   )
)

df_metrics.write.mode("overwrite").parquet(output_path)
```

7. Crear los Glue Jobs

Job 1

```
Name: bronze_to_silver_job
Runtime: Glue 4.0
```

```
Language: Python
Script: bronze_to_silver.py
Role: GlueDataLakeRole
Workers: G.1X
Number workers: 2
```

Job 2

```
Name: silver_to_gold_job
Runtime: Glue 4.0
Script: silver_to_gold.py
Role: GlueDataLakeRole
```

---

8. Ejecutar Pipeline

Orden de ejecución:

```
1 Upload CSV
2 Run bronze_to_silver_job
3 Run silver_to_gold_job
```

---

9. Resultado esperado

Silver (Parquet limpio)

```
s3://datalake-silver/sales/
```

Schema:

```
order_id
customer_id
product
amount
city
date
```

Gold (Dataset analítico)

```
s3://datalake-gold/sales_metrics/
```

Ejemplo:

```
city        total_sales    num_orders
Bogota         1295            3
Medellin       1175            2
Cali            300            1
```

---

10. Consulta desde Athena

Crear tabla externa:

```
CREATE EXTERNAL TABLE gold_sales_metrics (
   city string,
   total_sales double,
   num_orders int
)
STORED AS PARQUET
LOCATION 's3://datalake-gold-123456789/sales_metrics/';
```

Consulta:

```
SELECT * FROM gold_sales_metrics;
```

Costeo Glue

1. Precio oficial de Glue

El costo se calcula con **DPU-hour**.

Precio aproximado:

1 DPU-hour = $0.44 USD

Ejemplo oficial:

6 DPU
15 minutos

Costo = 6 × 0.25 h × 0.44
Costo = $0.66

2. Qué es un DPU

Un **DPU (Data Processing Unit)** tiene aproximadamente:

```
4 vCPU
16 GB RAM
```

3. Workers de Glue

Worker	DPU	vCPU	RAM
G.1X	1	4	16 GB
G.2X	2	8	32 GB
G.4X	4	16	64 GB

---

4. Fórmula de costo

La fórmula completa:

```
Costo = DPU × Tiempo_horas × Precio_DPU
```

Donde:

Precio_DPU ≈ 0.44 USD

---

5. Fórmula aproximada para estimar DPUs

Regla empírica usada en data engineering:

DPUs ≈ Tamaño_datos_GB / 10

Otra aproximación más segura:

DPUs ≈ Tamaño_datos_GB / 5

Depende de:

- joins
- aggregations
- skew
- tamaño de partición

6. Fórmula de tiempo aproximado

Para ETL simples (CSV → Parquet):

Tiempo ≈ Tamaño_GB / (DPU × 1.5)

7. Estimación para tu pipeline

Supongamos pipeline:

CSV → Parquet
Parquet → Aggregation

8. Tabla recomendada de configuración

Dataset pequeño

10 MB

Configuración:

```
Worker: G.1X
Workers: 2
DPUs = 2
Tiempo ≈ 2 min
```

Costo:

2 × (2/60) × 0.44= $0.029≈ **3 centavos**

Dataset pequeño realista

100 MB

Configuración:

```
Worker: G.1X
Workers: 2
DPUs = 2
Tiempo ≈ 3 min
```

Costo:

2 × (3/60) × 0.44= $0.044≈ **4 centavos**

Dataset mediano

1 GB

Configuración recomendada:

```
Worker: G.1X
Workers: 4
DPUs = 4
Tiempo ≈ 4 min
```

Costo:

4 × (4/60) × 0.44= $0.117≈ **12 centavos**

Dataset analítico

10 GB

Configuración:

```
Worker: G.1X
Workers: 8
DPUs = 8
Tiempo ≈ 6 min
```

Costo:

8 × (6/60) × 0.44= $0.352≈ **35 centavos**

Dataset grande

100 GB

Configuración:

```
Worker: G.2X
Workers: 10
DPUs = 20
Tiempo ≈ 10 min
```

Costo:

20 × (10/60) × 0.44= $1.46

Dataset muy grande

1 TB

Configuración:

```
Worker: G.4X
Workers: 20
DPUs = 80
Tiempo ≈ 20 min
```

Costo:

80 × (20/60) × 0.44 = $5.86

9. Tabla resumen

Data size	Worker	Workers	DPUs	Tiempo	Costo
10 MB	G.1X	2	2	2 min	$0.03
100 MB	G.1X	2	2	3 min	$0.04
1 GB	G.1X	4	4	4 min	$0.12
10 GB	G.1X	8	8	6 min	$0.35
100 GB	G.2X	10	20	10 min	$1.46
1 TB	G.4X	20	80	20 min	$5.86

State Machine (ASL) para ejecutar los 2 Glue Jobs

Esto usa la integración nativa **sync**, así Step Functions **espera a que cada job termine** antes de pasar al siguiente.

Escritura

```
{
"Comment": "Pipeline Bronze -> Silver -> Gold con Glue",
"StartAt": "bronzeToSilverJob",
"States": {
"bronzeToSilverJob": {
"Type": "Task",
"Resource": "arn:aws:states:::glue:startJobRun.sync",
"Parameters": {
"JobName": "bronze_to_silver"
},
"Next": "silverToGoldJob"
},
"silverToGoldJob": {
"Type": "Task",
"Resource": "arn:aws:states:::glue:startJobRun.sync",
"Parameters": {
"JobName": "silver_to_gold"
},
"End": true
}
}
}
```

Costo de Step Functions (la máquina de estados)

En el tipo más común (**Standard Workflow**):

- Pagas **por transición de estado** (cada paso ejecutado).
- Precio aproximado: **$0.000025 por transición**.
- Tienes **4,000 transiciones gratis al mes**.

Ejemplo simple:

State machine:

Start
-> Glue Job 1
-> Glue Job 2
-> Glue Job 3
-> End

Eso son aproximadamente **5 transiciones**.

Costo por ejecución:

5 * $0.000025 = $0.000125

Si ejecutas **1000 veces al mes**:

1000 * 5 = 5000 transiciones
5000 - 4000 gratis = 1000 pagas
1000 * 0.000025 ≈ $0.025

Role IAM para Step Functions

Debes crear un **IAM Role** que use la state machine.

Trust policy

Permite que Step Functions asuma el rol.

Escritura

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "Service": "states.amazonaws.com"
      },
      "Action": "sts:AssumeRole"
    }
  ]
}
```

Policy para ejecutar los Glue Jobs

Esto permite:

- iniciar jobs
- consultar estado

Escritura

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "glue:StartJobRun",
        "glue:GetJobRun",
        "glue:BatchStopJobRun"
      ],
      "Resource": [
        "arn:aws:glue:*:*:job/bronze_to_silver",
        "arn:aws:glue:*:*:job/silver_to_gold"
      ]
    }
  ]
}
```

Permisos para S3 (si la state machine necesita leer inputs)

Normalmente **Glue jobs** tienen su propio rol con acceso a **Amazon S3**.

Ejemplo para los buckets:

Escritura

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "s3:ListBucket"
            ],
            "Resource": [
                "arn:aws:s3:::datalake-bronze-714647503442",
                "arn:aws:s3:::datalake-silver-714647503442",
```

```
                "arn:aws:s3:::datalake-gold-714647503442",
                "arn:aws:s3:::aws-glue-assets-714647503442-us-east-1"
            ]
        },
        {
            "Effect": "Allow",
            "Action": [
                "s3:GetObject",
                "s3:PutObject",
                "s3:DeleteObject"
            ],
            "Resource": [
                "arn:aws:s3:::datalake-bronze-714647503442/*",
                "arn:aws:s3:::datalake-silver-714647503442/*",
                "arn:aws:s3:::datalake-gold-714647503442/*",
                "arn:aws:s3:::aws-glue-assets-714647503442-us-east-1/*"
            ]
        }
    ]
}
```

Esto normalmente va en el **IAM role de Glue**, no en Step Functions.